

LECTURE 4

Multivariate Dynamics: VAR, Cointegration and VEC

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Agenda

- 01 Vector Autoregression: specification and estimation
- 02 Granger causality and lag selection
- 03 Impulse response functions
- 04 Variance decomposition
- 05 Structural VAR identification
- 06 Johansen cointegration framework
- 07 Vector Error Correction Model (VEC)
- 08 Business applications

(i) Macro-finance: US federal funds rate, output gap, and inflation (monetary policy analysis). (ii) Marketing: advertising, sales, and price for a consumer goods brand.

Section 1

Vector Autoregression

The VAR(p) Model I

$$X_t = \mu + A_1 X_{t-1} + \cdots + A_p X_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \Sigma)$$

X_t : $n \times 1$ vector of endogenous variables; A_i : $n \times n$ coefficient matrices; Σ : $n \times n$ positive definite covariance matrix.

Key features

- Each equation is a reduced-form regression of $X_{i,t}$ on lags of **all** variables
- Estimation: OLS equation-by-equation (efficient when regressors are the same)
- No a priori restrictions on who causes whom
- Sims (1980): alternative to large structural models with many debatable exclusion restrictions

The VAR(p) Model II

The VAR(p) is stationary if all eigenvalues of the companion matrix lie **inside** the unit circle. Equivalently, all roots of $\det(I - A_1z - \dots - A_pz^p) = 0$ lie **outside** the unit circle.

Test: check the modulus of all eigenvalues. Plot roots in the complex unit disc.

The VAR(p) Model III

Lag length selection

Too few lags $\Rightarrow$ serially correlated residuals, invalid inference. Too many lags $\Rightarrow$ parameter proliferation, poor forecasts.

Criterion	Formula	Property
AIC	$\ln \hat{\Sigma} + 2pn^2/T$	Not consistent; tends to overfit
BIC	$\ln \hat{\Sigma} + pn^2 \ln T/T$	Consistent; preferred
HQIC	$\ln \hat{\Sigma} + 2pn^2 \ln \ln T/T$	Consistent, intermediate penalty

Always check residuals for serial correlation after selecting p (Portmanteau test or LM test for VAR residuals). Also check for normality and ARCH effects.

Section 2

Granger Causality, IRF and Variance Decomposition

In a bivariate VAR(p) for (Y_t, Z_t) : Z_t does **not Granger-cause** Y_t if all lags of Z are jointly zero in the Y equation:

$$H_0 : A_{12,1} = A_{12,2} = \cdots = A_{12,p} = 0$$

Test: standard F -test (or χ^2 Wald test). Degrees of freedom: p restrictions.

Interpretation

- Granger causality is a **predictive** concept: does past Z improve forecasts of Y beyond what past Y alone provides?
- It does **not** imply structural (economic) causation
- Test is valid for stationary VAR; requires modification for $I(1)$ variables (Toda–Yamamoto procedure)

Advertising–sales: Does past advertising Granger-cause sales? Does past sales Granger-cause advertising (feedback)?

Dekimpe & Hanssens (1995): in many categories, both directions are significant — advertising and sales are jointly determined.

Impulse Response Functions (IRFs)

How does $X_{j,t+h}$ respond to a one-unit shock in $X_{i,t}$, holding all else constant?

MA(∞) representation: $X_t = \mu + \sum_{h=0}^{\infty} \Psi_h \varepsilon_{t-h}$

$$\text{IRF}_{ij}(h) = \frac{\partial X_{j,t+h}}{\partial \varepsilon_{i,t}} = [\Psi_h]_{ji}$$

- Requires **identification** of shocks: ε_t are correlated, so a “unit shock to ε_i ” must be defined
- **Cholesky decomposition**: $\Sigma = PP'$, orthogonalise with $u_t = P^{-1}\varepsilon_t$
- Cholesky ordering matters: variable listed first is causally prior; not always credible
- **Forecast error variance decomposition (FEVD)**: how much of the h -step forecast variance of X_j is attributable to shocks in X_i ?

IRFs are estimated with substantial uncertainty. Always plot bootstrap 68% or 90% confidence bands. Bands that include zero imply the response is not statistically distinguishable from zero.

Section 3

Structural VAR

SVAR Identification I

Reduced form VAR: $X_t = \mu + A_1 X_{t-1} + \dots + \varepsilon_t$, Σ unrestricted.

Structural form: $B_0 X_t = \nu + B_1 X_{t-1} + \dots + e_t$, $e_t \sim (0, I)$.

The structural shocks e_t are the economically meaningful disturbances (demand shock, supply shock, monetary shock, etc.).

Cholesky (recursive) identification

- B_0 is lower triangular
- Variable i does not respond contemporaneously to shocks in variables $j > i$
- Requires credible causal ordering
- Simple and widely used but often ad hoc

Sign restrictions (Uhlig, 2005)

- Do not impose exact zero restrictions
- Instead, restrict signs of IRFs (e.g., a monetary tightening raises rates and reduces output)
- More agnostic identification
- Yields set of admissible models (identified set)

Kilian (2009) oil market SVAR — a classic example

Three-variable SVAR: global oil supply (Δprod_t), global economic activity (rea_t), real oil price (rpo_t).

Identification: recursive ordering with supply shock first, then demand shock.

- Oil supply shock: immediate effect on supply, no same-period output response
- Aggregate demand shock: affects both global activity and oil prices contemporaneously
- Oil-specific demand shock: residual — precautionary demand

SVAR is the standard tool for measuring how commodity price shocks transmit to firm revenues, input costs, and consumer demand. The Kilian framework is directly applicable to energy-intensive industries.

Section 4

Johansen Cointegration and VEC

The Johansen Framework I

From VAR to VEC: the key step

A VAR(p) in levels can be reparameterised as:

$$\Delta X_t = \mu + \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \varepsilon_t$$

where $\Pi = \sum_{i=1}^p A_i - I$ and $\Gamma_i = -\sum_{j=i+1}^p A_j$.

The rank of Π determines the cointegration structure:

- $\text{rank}(\Pi) = 0$: no cointegration, $X_t \sim I(1)$ with no long-run equilibrium $\Rightarrow$ VAR in differences
- $\text{rank}(\Pi) = n$: all variables stationary $\Rightarrow$ VAR in levels
- $0 < \text{rank}(\Pi) = r < n$: **r cointegrating relations** $\Rightarrow$ VEC model

When $r = \text{rank}(\Pi)$: $\Pi = \alpha\beta'$ where β ($n \times r$): cointegrating vectors; α ($n \times r$): adjustment (loading) matrix.

Johansen trace and maximum eigenvalue tests

H_0 : at most r cointegrating vectors vs. H_1 : more than r :

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i)$$

Test sequentially: $r = 0, 1, 2, \dots$. Stop when H_0 is not rejected.

H_0 : exactly r cointegrating vectors vs. H_1 : $r + 1$:

$$\lambda_{\text{max}}(r, r + 1) = -T \ln(1 - \hat{\lambda}_{r+1})$$

Five cases for deterministic components (no constant, restricted constant, unrestricted constant, restricted trend, unrestricted trend). Choice affects critical values. Match to the data — a restricted constant (case 2) is most common for $I(1)$ variables with non-zero mean.

The Johansen Framework IV

Identifying the cointegrating vectors

When $r > 1$, the cointegrating space is identified but individual vectors are not — need normalisation restrictions.

Common approaches:

- **Normalise one coefficient per vector** (e.g., set $\beta_{11} = 1$)
- **Economic theory restrictions**: long-run homogeneity, unit elasticities
- **Exclusion restrictions**: set some $\beta_{ij} = 0$ (zero restrictions on the cointegrating space)

For two stock prices (P_{1t}, P_{2t}) both $I(1)$: test $r = 1$ cointegrating vector $[1, -\theta]$. The spread $P_{1t} - \theta P_{2t}$ is stationary — it is the basis for statistical arbitrage. The VEC model governs how each price adjusts to restore equilibrium.

Applications

- **Price discovery:** VAR/VEC between spot and futures prices; which market leads price discovery?
- **Macro-finance linkages:** SVAR with interest rates, inflation, output gap for monetary policy
- **Advertising-sales dynamics:** Bivariate VAR to test Granger causality and compute IRFs
- **Supply chain:** Multivariate VAR to trace propagation of commodity price shocks
- **Pairs trading:** VEC framework for constructing stationary spread strategies

Key references

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- Bass, F.M. et al. (2007). Wearout effects of advertising. *Marketing Science*, 26, 179–195.

Software: R (vars: VAR, VECM; urca: ca.jo) or Python (statsmodels.tsa.vector_ar). Applied exercises: estimate a bivariate VAR for advertising–sales; test Granger causality; plot IRFs with bootstrap bands; run Johansen test.

Next: Lecture 5

Structural Breaks, State Space and the Kalman Filter

Chow · Bai–Perron · Kalman filter · Time-varying parameters